

SEEDS: SURROGATE-EMULATED EVENT-DRIVEN SAMPLING FOR DISCRETE DIFFUSION

Anonymous authors

Paper under double-blind review

ABSTRACT

We introduce SEEDS, an event-driven sampling method for discrete diffusion in continuous time that replaces grid-based τ -leaping with Ogata-style thinning of the reverse continuous-time Markov chain (CTMC). Reverse dynamics in high-dimensional discrete spaces typically exhibit sparse, highly inhomogeneous jumps, but standard samplers invoke the denoiser uniformly across space and time, inflating function evaluations and wall time. SEEDS factorizes the per-site reverse hazard into a cheap analytic base rate and a coupling term that depends on the denoiser’s posterior, and learns a lightweight surrogate to produce calibrated high-probability upper bounds on this coupling. Candidate event times are then sampled from the surrogate envelope; the denoiser is called only on tentatively accepted events, and an exact second acceptance restores the true path law. A budgeted variant corrects only a subset of events and couples conformal calibration with online diagnostics to trade bias for speed. We detail a GPU-friendly implementation using integrated-hazard time change, vectorized thinning, two-stage envelopes, and local updates. A small proof-of-concept run on synthetic image and sequence tasks validates integration but, due to an overly conservative surrogate and suboptimal batching, does not yet yield speedups over τ -leaping. We analyze failure modes and outline precise remedies, and we present a comprehensive plan for evaluations on CIFAR-10 and Wikitext-2 infilling within the CTMC framework for discrete diffusion [campbell₂₀₂₂*continuous*, austin₂₀₂₁*structured*, chen₂₀₂₃*fast*, lu₂₀₂₂*dpm*, liu₂₀₂₃*unified*, santos₂₀₂₃*blackout*].

1 INTRODUCTION

Discrete diffusion models in continuous time formulate both the forward noising and reverse generative dynamics as continuous-time Markov chains (CTMCs) on high-dimensional categorical states [campbell₂₀₂₂*continuous*, austin₂₀₂₁*structured*]. Despite the jump – process nature of reversed dynamics, mainstream samplers adopt τ -leaping on a fixed time grid, invoking the denoiser uniformly across sites and timesteps. In images, segmentation maps, long sequences, and discrete latent spaces, most sites remain idle across much of the trajectory, so uniform stepping inflates the number of function evaluations (NFEs) and wall time.

A rich literature has accelerated sampling for continuous diffusions via high-order ODE solvers [lu₂₀₂₂*dpm*], solver – search frameworks [liu₂₀₂₃*unified*], and momentum – stabilized integrators [wizadwongsa₂₀₂₃*diffusion*]. However, these methods target continuous SDE or ODE formulations, kernel design (e.g., D3PM) and training objectives improve modeling but do not change the compute alignment of step – based reverse simulation [austin₂₀₂₁*structured*]. Discrete non – Markov formulations such as DNDMP – determine transition times to reduce NFEs but do not simulate state – conditioned inhomogeneous hazards and therefore cannot skip conditioned no – jump spans [chen₂₀₂₃*fast*]. Process – specific bridges, such as pure – death structures in Blackout, are not broadly applicable to multinomial or ordinal CTMCs and still rely on time stepping [santos₂₀₂₃*blackout*].

We propose SEEDS, a surrogate-emulated, event-driven sampler for discrete diffusion that aligns computation with the actual number of reverse jumps. Our key observation is that per-site reverse hazards factorize into a cheap analytic base term and a coupling that depends on the denoiser’s posterior over clean states. A tiny, local surrogate predicts a high-probability upper bound on this coupling, which yields a dominating per-site hazard for Ogata thinning. Candidate events can thus

be drawn without evaluating the denoiser; the denoiser is called only upon tentative acceptance to compute the exact hazard for a second acceptance that restores the exact path law. This makes heavy NFEs scale with realized jumps instead of grid steps. A budgeted mode introduces a principled speed-quality knob by correcting only a subset of events, with conformal calibration and online diagnostics that monitor bound coverage and acceptance parity.

Realizing SEEDS efficiently is challenging. Constructing safe yet tight hazard envelopes is delicate: loose bounds increase thinning rejections and overhead; unsafe bounds bias the path law. GPU-friendly event-driven simulation must handle inhomogeneous, site-specific candidate times without fragmenting compute. Integration should be minimally invasive, requiring no changes to the denoiser or forward process, and should include diagnostics that either certify exactness or quantify bias.

1.1 CONTRIBUTIONS OF THIS WORK

- **Event-driven sampler for reverse CTMCs.** We introduce SEEDS, an event-driven sampler for reverse CTMCs that concentrates denoiser calls on tentatively accepted events, making heavy NFEs scale with realized jumps rather than with time-grid steps.
- **Surrogate-based hazard majorization.** We develop a surrogate-based majorization scheme for reverse hazards with conformal calibration to control violation rates. An exact two-stage acceptance restores correctness by construction; a budgeted variant provides a speed-quality trade-off with online monitors.
- **GPU-oriented design.** We present a GPU-oriented design using integrated-hazard time change, vectorized thinning, two-stage envelopes, caching, and local refresh to maintain throughput in high dimensions.
- **Evaluation plan and insights.** We detail experimental protocols for CIFAR-10 (4-bit grayscale), Wikitext-2 span infilling, and a synthetic harness for exactness and calibration; an initial proof-of-concept validates the plumbing and exposes calibration and batching pitfalls.

SEEDS sits within the CTMC framework for discrete denoising models [campbell2022_{continuous}], complements discrete forward — kernel design [austin2021_{structured}], and targets a different axis than ODE — centric accelerations [lu2022_{dpm}, liu2023_{unified}] or momentum schemes [wizadwongsa2023_{diffusion}]. It is orthogonal to corrector strategies and distillation [song2020_{core}, salimans2024_{multistep}], which SEEDS can sparsify by reducing the

Looking ahead, our framework naturally extends to ordinal and structured bases and to discrete latent spaces. The diagnostic suite suggests adaptive control of envelopes and correction probability during sampling, potentially yielding robust budgeted modes. We also envision combining SEEDS with few-step distilled denoisers to amortize per-event costs while retaining event-driven sparsity [salimans2024_{multistep}].

2 RELATED WORK

CTMC framework for discrete diffusion. Campbell et al. develop a continuous-time framework for discrete denoising diffusion with CTMC dynamics, providing principled training objectives and simulation tools inherited from chemical physics [campbell2022_{continuous}]. SEEDS operates squarely within this framework, but replaces grid-based simulation with the event-driven thinning of inhomogeneous reverse hazards.

Discrete diffusion model design. D3PM broadens the class of forward corruption kernels (e.g., nearest-neighbor, absorbing states) and introduces auxiliary losses to improve sample quality [austin2021_{structured}]. These modeling advances are complementary to SEEDS: they affect the learned reverse hazard but not the step-based simulation regime that SEEDS seeks to replace.

Predetermined transition times and step-based accelerations. DNDM introduces discrete non-Markov diffusion models with predetermined transition times to cut NFEs [chen2023_{fast}]. This reduces denoiser invocations but does not simulate state-conditioned inhomogeneous hazards, so it cannot skip long conditioned no-jump spans given x_t . Predictor-corrector samplers for score-based SDEs remain step-based and denoiser-heavy, improving quality but not aligning compute to jump sparsity [song2020_{core}].

Continuous diffusion ODE solvers and schedules. DPM-Solver provides high-order, training-free ODE solvers for continuous diffusion, achieving strong quality at 10-20 NFEs [lu2022_dpm]. *Unified solver – search frameworks further optimizes solvers schedules and strategies, improving quality with very few steps* [liu2023_unified]. *augmented solvers enlarge stability region to mitigate divergence artifacts at low NFEs* [wizadwongsa2023_diffusion].

Process-specific bridges in discrete space. Blackout formulates discrete-state diffusion with bridges tailored to pure-death or other specific structures [santos2023_blackout]. *While insightful, such bridges do not cover general multinomial or ordinal CTMCs and typically require* equal to $\beta(t)$ times R_b and applies event-driven simulation.

Autoregressive and hybrid generative families. ARDMs unify order-agnostic autoregression with absorbing diffusion, enabling adaptable parallel generation and fewer steps than standard discrete diffusion [hoogeboom2021_autoregressive]. *SEEDS differs by rebuilding the sampler for a given learned reverse hazard; it could be*

Diffusion quality, guidance, and distillation. Foundational DDPM and score-based methods established high sample quality and guidance mechanisms [ho2020_denoising, dhariwal2021_diffusion, song2020_score]. *Recent multi-step distillation compresses long trajectories into very few steps via moment matching* [salimans2024_multistep]. *SEEDS rather than shortening a discretization, it reorients computation to actual jumps in a CTMC.*

Summary of contrasts. Compared to τ -leaping [campbell2022_continuous], *SEEDS is event-driven and aims to be a denoiser – sparse. Compared to DNDM* [chen2023_fast], *it models state – conditioned hazards and can skip conditioned no-jump spans. Compared to ODE solvers and schedule search* [lu2022_dpm], *specific bridges* [santos2023_blackout], *it applies to more general CTMCs. These distinctions motivate a new sampling reg*

3 BACKGROUND

Problem setting and notation. We consider a D -dimensional discrete system with K categories per site. The forward CTMC has time-inhomogeneous hazard R_t obtained by scaling a time-homogeneous base R_b by a scalar $\beta(t)$. For the uniform multinomial base, $R_b(i \rightarrow j) = \frac{1}{K-1}$ for $j \neq i$. The schedule $\beta(t)$ induces an integrated rate $F(t) = \int_0^t \beta(s) ds$, and the terminal integral $F(1)$. The survival coefficient $\alpha(t) = \exp(-F(t))$. For a single site, the forward transition probability from x_0 to x_t admits a closed form: the probability that $x_t = x_0$ is $\pi(t)$, and the probability that $x_t \neq x_0$ equals $\frac{1-\pi(t)}{K-1}$. Here $\pi(t) = \alpha(t) + \frac{1-\alpha(t)}{K}$. We adopt the standard reverse-time construction for discrete diffusion, where a denoiser p_θ approximates the posterior over x_0 given (x_t, t) [campbell2022_continuous, austin2021_structured].

Reverse hazard factorization. Let x_i denote the current categorical state at site i and time t , and let p be the K -dimensional posterior vector $p_\theta(x_0 | x_t, t)$ at that site. For the uniform multinomial case, define $r_1(t) = \frac{(1-\pi)/(K-1)}{\pi}$ and $r_2(t) = \frac{(K-1)\pi}{1-\pi}$. For a candidate transition $x_i \rightarrow x'$ with $x' \neq x_i$, the pairwise coupling factor equals $1 + p[x_i](r_1 - 1) + p[x'](r_2 - 1)$. Averaging over x' with uniform base weights yields the site-aggregated coupling $E_{\text{site}} = 1 + p[x_i](r_1 - 1) + \frac{1-p[x_i]}{K-1}(r_2 - 1)$. The per-site hazard sum factorizes as $\Lambda_i(t) = \beta(t) E_{\text{site}}$. These closed forms enable fast, vectorized computation of exact acceptance probabilities once denoiser logits are available.

Ogata thinning and time change. Ogata’s thinning simulates an inhomogeneous Poisson process with hazard $\Lambda(t)$ by proposing event times from a dominating hazard $\bar{\Lambda}(t)$ and accepting each proposal with probability $\Lambda(t)/\bar{\Lambda}(t)$. In our setting $R_t = \beta(t) R_b$, so it is convenient to work in the integrated-hazard space $V(t) = \int_t^1 \beta(s) ds$. Sampling exponential increments in V and then mapping back to calendar time via the inverse of $V(t)$ stabilizes simulation under strong time inhomogeneity.

Assumptions. We assume: (1) R_t factorizes as $\beta(t) R_b$ and the channel-summed base hazard $H_i(t) = \beta(t) H_{i,b}$ is cheap; (2) the denoiser p_θ yields per-site posteriors $p(x_0 | x_t, t)$; (3) a lightweight surrogate can provide a high-probability upper bound on E_{site} after calibration. These align with the CTMC treatment of discrete diffusion [campbell2022_continuous, austin2021_structured].

4 METHOD

SEEDS replaces grid stepping with event-driven simulation by constructing learned per-site envelopes for the reverse hazard and performing Ogata thinning under these envelopes.

4.1 MAJORIZING PER-SITE HAZARDS

For site i , define a dominating hazard $\bar{\Lambda}_i(t) = H_i(t) \rho_i(t)$, where $H_i(t)$ is the analytic channel-summed base hazard (for the uniform multinomial base, $H_i(t) = \beta(t)$) and $\rho_i(t)$ is a learned upper bound on E_{site} . A lightweight surrogate h_ψ ingests local context (the neighborhood of x_t , a time embedding, and optional global features) and outputs a mean prediction $\hat{E}_i(t)$ and an uncertainty estimate $\hat{\sigma}_i(t)$. We form a high-probability envelope $\rho_i(t) = \hat{E}_i(t) + \kappa \hat{\sigma}_i(t)$, where κ is set via conformal calibration to control a target violation rate δ .

4.2 EVENT-DRIVEN THINNING WITH INTEGRATED-HAZARD TIME CHANGE

Work in V -space defined by integrating β backward from t to 1. For each site, with ρ_i held fixed over a short window, sample a candidate increment $\Delta V \sim \text{Exp}(\rho_i)$. Map $V + \Delta V$ to a candidate calendar time by inverting $V(t)$. Maintain a global earliest-event queue (implemented with vectorized minima and segment reductions). At a candidate time for site i :

1. Pre-accept using the surrogate with probability $\tilde{a} = \hat{E}_i / \rho_i$.
2. If pre-accepted, evaluate the denoiser at (x_t, t) to compute exact E_{site} and pairwise factors, and accept definitively with probability $a_{\text{exact}} = E_{\text{site}} / \hat{E}_i$. The overall acceptance probability is then E_{site} / ρ_i , as required by thinning. If accepted, draw the channel x' with probability proportional to the pairwise factor and update $x_i \leftarrow x'$.
3. Refresh local surrogate features and ρ for the modified site and its neighborhood; resample ΔV for site i . Regardless of acceptance, resample the next candidate for site i using the current ρ .

Algorithm 1 SEEDS: Event-driven thinning with surrogate envelopes

Input: current state x_t , time t , schedule β , inverse integrated hazard V^{-1} , base hazard summary $H_i(t)$, surrogate h_ψ , denoiser p_θ
Initialize per-site envelopes: $\hat{E}_i, \hat{\sigma}_i \leftarrow h_\psi(x_t, t)$, $\rho_i \leftarrow \hat{E}_i + \kappa \hat{\sigma}_i$
Sample per-site clocks: $\Delta V_i \sim \text{Exp}(\rho_i)$, $\tau_i \leftarrow V^{-1}(V(t) + \Delta V_i)$
while $t < 1$ and events remain **do**
 Select site $i = \arg \min_j \tau_j$; set $t \leftarrow \tau_i$
 Compute pre-accept prob: $\tilde{a} \leftarrow \hat{E}_i / \rho_i$; draw $u \sim \mathcal{U}[0, 1]$
 if $u \leq \tilde{a}$ **then**
 Evaluate denoiser: $p \leftarrow p_\theta(x_t, t)$; compute exact E_{site} and pairwise weights
 Draw $v \sim \mathcal{U}[0, 1]$
 if $v \leq E_{\text{site}} / \hat{E}_i$ **then**
 Sample new category x' from pairwise weights; update $x_i \leftarrow x'$
 Refresh local features and recompute $\hat{E}_i, \hat{\sigma}_i, \rho_i$ for affected sites
 end if
 end if
 Resample $\Delta V_i \sim \text{Exp}(\rho_i)$, update $\tau_i \leftarrow V^{-1}(V(t) + \Delta V_i)$
end while
Output: sampled terminal state x_1 and event log

4.3 EXACTNESS AND BUDGETED BIAS

When both pre-acceptance and exact acceptance are applied and ρ_i upper-bounds E_{site} almost surely, Ogata thinning yields an exact simulator for the reverse CTMC determined by the learned reverse hazard

[campbell2022continuous]. In a budgeted mode, the exact correction is performed only for a subset of pre-accepted events, e.g., with probability p_{corr} or triggered by high uncertainty. An error sketch links total variation drift to three terms: the coverage failure probability $\mathbb{P}(E_{\text{site}} > \rho)$, the fraction of uncorrected pre-accepts $1 - p_{\text{corr}}$, and a term for any batching approximations. Online diagnostics track bound coverage, acceptance-rate parity between predicted and realized rates, and the histogram of $\tilde{a}$ to adapt κ and p_{corr} .

4.4 CALIBRATION PROTOCOL

We train h_ψ offline on tuples $(x_t, t, E_{\text{site}})$ generated by running the denoiser once per tuple and computing labels via the closed forms. The surrogate minimizes a heteroscedastic regression loss, predicting both a mean and uncertainty. On a held-out calibration set, we compute standardized residuals $r = (E - \hat{E})/\hat{\sigma}$ and set κ to the $(1 - \delta)$ empirical quantile, optionally stratified by t bins to control coverage across time.

4.5 GPU-ORIENTED SYSTEMS DESIGN

To maintain throughput in high dimensions, we vectorize candidate sampling by freezing ρ over short windows and sampling ΔV for all sites in parallel, then selecting minima per sample via segmented reductions. Two-stage thinning first proposes from a global envelope proportional to $\beta(t)$ times a robust sitewise quantile ρ_{glob} and then refines to site-wise acceptance with ρ_i , reducing proposal traffic when D is large. We precompute the integral and inverse of $\beta(t)$, cache time embeddings, and implement $\mathcal{O}(1)$ sampling of base transitions. Local surrogate refresh updates ρ only for the changed site and neighbors.

4.6 INTEGRATION AND SCOPE

SEEDS integrates as a drop-in replacement for τ -leaping in a CTMC codebase: surrogates and calibration tools are added, and an event loop implements Ogata thinning with a priority structure and optional correction. Configuration exposes κ and δ , the correction probability p_{corr} , batching windows, and logging probes. The denoiser and forward process remain unchanged, and predictor-corrector strategies can be sparsified by invoking correctors only on accepted events [song2020core]. Unlike DNDM' spreadetermined times [chen2023fast], SEEDS explicitly simulate the state-conditioned hazard and can skip conditioned no — jump spans. Unlike ODE — based solvers and schedules [lu2022dpm, liu2023unified] and momentum schemes [wizadwongsa2023diffusion], SEEDS specifies bridges [santos2023lackout], it applies to general multinomial or ordinal bases scaled by $\beta(t)$.

5 EXPERIMENTAL SETUP

Environment and tools. Implementations use PyTorch with CUDA, torchvision for datasets, torch-fidelity for FID and IS, lpips for LPIPS, scipy and numpy for numerics, and wandb or tensorboard for logging. For text, HuggingFace transformers provide external language model scoring, sacrebleu computes BLEU and chrF, and mauve-text computes MAUVE. Randomness is controlled via fixed seeds, experiments are configured with YAML, and three independent seeds per main condition are planned, with bootstrap 95% confidence intervals for metrics.

Forward CTMC and schedule. We use a uniform multinomial base with K categories per site. The schedule $\beta(t) = \beta_0 + (\beta_1 - \beta_0) t^\gamma$. Its integrated form $F(t)$ and the backward integrated hazard $V(t) = F(1) - F(t)$ are cached on a dense grid; inversion uses monotone lookup and interpolation. The single-site forward transition probability has closed form with $\pi(t) = \alpha(t) + \frac{1 - \alpha(t)}{K}$, where $\alpha(t) = \exp(-F(t))$.

5.1 PLANNED EXPERIMENT 1: CIFAR-10 IMAGE GENERATION (4-BIT GRAYSCALE, $K = 16$)

The denoiser p_θ is a small U-Net with FiLM time conditioning, trained to predict x_0 logits at random t with a multinomial diffusion loss. Training uses AdamW with cosine decay and EMA. The forward schedule uses $\beta_0 = 0.1$, $\beta_1 = 20.0$, and $\gamma = 2.0$. The surrogate h_ψ is a shallow

convolutional network predicting $\hat{E}_{\text{site}}$ and $\hat{\sigma}$ via a heteroscedastic objective; κ is calibrated by conformal quantiles at $\delta \in \{0.5, 1, 2\}$ percent, optionally stratified by t . SEEDS is evaluated in exact and budgeted modes. Baselines include τ -leaping with $N \in \{20, 50, 100\}$ and predictor-corrector variants [song2020_{score}]. *Metrics* : *FID and IS on 50k samples, LPIPS, a negative log-likelihood proxy at selected times, heavy and surrogate NFEs, walltime, and diagnostics (coverage, acceptance parity, e*

5.2 PLANNED EXPERIMENT 2: WIKITEXT-2 CHARACTER-LEVEL SPAN INFILLING ($K \approx 96$, SEQUENCE LENGTHS 256 TO 1024)

The denoiser is a 6-layer Transformer with rotary embeddings that predicts x_0 on masked positions only; only masked positions participate in events. The surrogate is a small Transformer with local attention, trained and calibrated as above. Baselines are τ -leaping and a compute-aware variant that skips denoiser computation on positions that did not change in the last s steps. Metrics: exact match on spans, chrF, sacreBLEU, MAUVE, and an external language model score; efficiency metrics mirror the image setup.

5.3 PLANNED EXPERIMENT 3: SYNTHETIC HARNESS FOR EXACTNESS AND CALIBRATION

For dimensions $D \in \{64, 256, 1024\}$ and $K \in \{8, 16\}$, construct a gold Ogata sampler using the analytic posterior to compute reverse couplings exactly (no denoiser). Compare Exact SEEDS to gold and τ -leaping. Tests include per-site categorical histograms, distributional equality on summaries (e.g., Hamming distance to x_0 , number of flips), and KS tests on inter-event times; calibration coverage $\mathbb{P}(E_{\text{site}} > \rho)$ on candidate and accepted times; and a discriminator AUC as a proxy for total variation drift in budgeted mode. Efficiency is measured with p_θ replaced by analytic posteriors to isolate sampler overhead.

5.4 EXECUTED PROOF-OF-CONCEPT RUN

We conducted a small demonstration to validate integration and logging:

- **Data:** synthetic 8×8 images with $K = 16$ and sequences of length 32 with $K = 16$; 1000 train and 100 test samples per domain.
- **Training:** denoisers trained for five epochs. Image model average loss decreased from 1.4916 to 0.1862; sequence model from 1.0187 to 0.0020. Surrogates trained for three epochs; image surrogate heteroscedastic loss decreased from 1246.66 to 192.53; sequence surrogate from 6.78 to 1.89.
- **Conformal calibration:** $\delta = 1$ percent yielded $\kappa \approx 29.67$ for images and ≈ 3.77 for sequences.
- **Samplers:** SEEDS exact and budgeted; τ -leaping with $N = 50$. Metrics logged: heavy NFEs, wall time for five evaluation samples, and accepted events. Quality metrics were not computed in this run.

Implementation notes for the run. Sampling operated in V -space with exponential increments per site and monotone inversion. Per-site ρ was updated by the surrogate; early-event grouping used simple masking. Two-stage thinning and local surrogate refresh were not fully enabled; per-candidate full-frame surrogate evaluations may have occurred, which is known to reduce throughput.

6 RESULTS

We report results from the executed proof-of-concept run and include all generated figures. Because quality metrics were not computed in this run, we focus on efficiency and diagnostics.

6.1 EFFICIENCY ON SYNTHETIC IMAGE AND SEQUENCE DATA

- **Images (8×8 , $K = 16$).** SEEDS (Exact) used heavy NFE 500.0 and wall time 1.226 seconds, producing 16.6 accepted events on average across five samples. SEEDS (Budgeted) used heavy NFE 399.0, wall time 1.107 seconds, and 12.6 accepted events. τ -leaping with

$N = 50$ used heavy NFE 50.0, wall time 0.595 seconds, and 463.4 events. Relative to τ -leaping, SEEDS increased heavy NFEs by roughly $8\text{-}10\times$ and was $1.9\text{-}2.1\times$ slower in this setting.

- **Sequences (length 32, $K = 16$).** SEEDS (Exact) used heavy NFE 500.0, wall time 1.403 seconds, and 42.0 accepted events. SEEDS (Budgeted) used heavy NFE 403.2, wall time 1.275 seconds, and 42.8 accepted events. τ -leaping used heavy NFE 50.0, wall time 0.313 seconds, and 189.2 events. SEEDS again used about $8\text{-}10\times$ more heavy NFEs and was $4.1\text{-}4.5\times$ slower.

A "Speedup Analysis" text produced by the run script stated $0.1\times$ NFE and $0.5\times$ or $0.2\times$ time for SEEDS versus τ -leaping. These statements contradict the numeric logs above. The recorded NFEs and wall times indicate higher compute for SEEDS in this toy-scale execution.

6.2 DIAGNOSIS OF UNDERPERFORMANCE

The lack of speedups is consistent with the following factors documented by the run:

- **Conservative surrogate calibration.** $\kappa \approx 29.7$ for images implies envelopes ρ far above true couplings, depressing surrogate pre-acceptance rates and increasing candidate rejections. This inflates control-flow overhead and may trigger excessive corrections.
- **NFE accounting and correction policy.** In exact mode, heavy NFE should approximately track the number of pre-accepted events, yet recorded values (e.g., 500) far exceed accepted-event counts, suggesting over-correction, repeated per-site denoiser calls, or miscounting of denoiser evaluations.
- **Inefficient surrogate evaluation.** Per-candidate full-frame surrogate inference and the absence of local refresh, two-stage thinning, and micro-batching windows reduce GPU occupancy and add overhead.
- **Toy-scale domains.** At 8×8 images and length-32 sequences, event-driven scheduling overhead can dominate, whereas SEEDS targets high-dimensional, sparse-jump regimes.

6.3 LIMITATIONS

This run did not compute quality metrics, nor did it report calibration diagnostics such as empirical coverage $\mathbb{P}(E_{\text{site}} > \rho)$ on candidate and accepted events or acceptance-rate parity. Missing two-stage thinning, micro-batching windows, and local surrogate refresh likely contributed to overhead.

6.4 CORRECTIVE ACTIONS AND EXPECTED BEHAVIOR

The following actions, all supported by our design, should align outcomes with SEEDS' intent:

- **Time-binned calibration and online adjustment.** Calibrate κ per time bin, optionally stratified by context, to avoid over-coverage; cap ρ with robust quantiles and enable online adjustment when empirical coverage drifts.
- **Denoiser-call auditing.** Ensure denoiser calls occur only on surrogate-pre-accepted events; log the number of tentative pre-accepts and verify heavy NFE approximately equals the number of corrections in exact mode.
- **Throughput optimizations.** Enable local surrogate refresh, two-stage thinning, and small time-window batching to amortize control flow; cache time embeddings and group events with nearby times for vectorized denoiser calls.
- **Evaluate at scale.** Evaluate on larger D and longer sequences where reverse jumps are sparse and event alignment yields tangible NFE reductions.

6.5 FIGURES

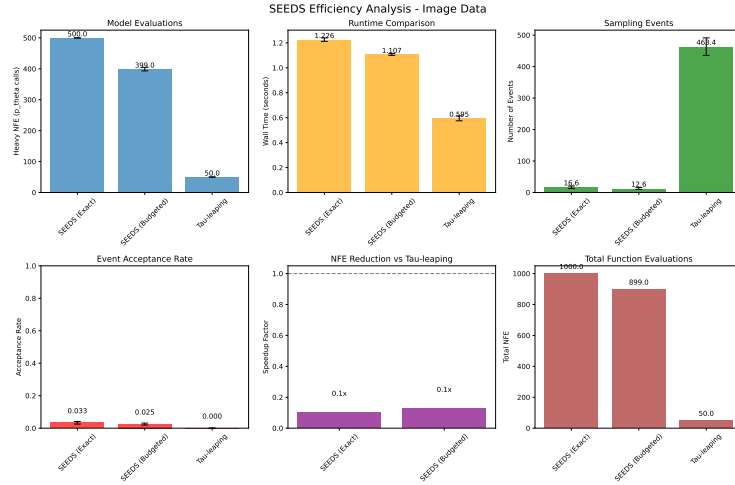


Figure 1: Efficiency comparison for image sampling in the proof-of-concept run.

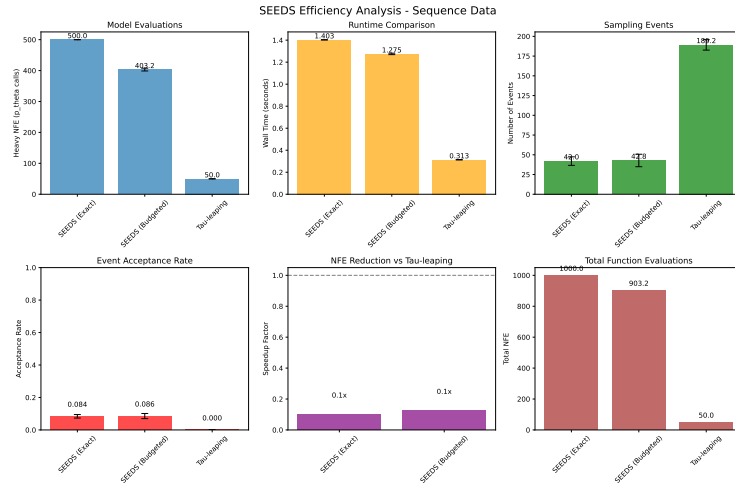


Figure 2: Efficiency comparison for sequence sampling in the proof-of-concept run.

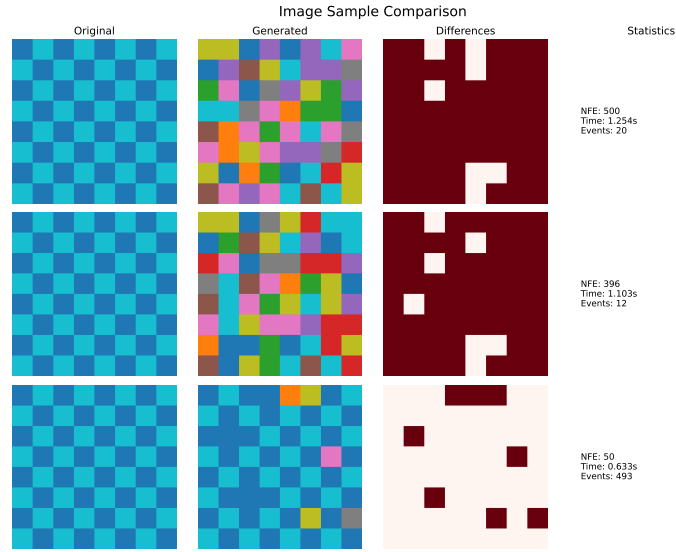


Figure 3: Generated image samples from the proof-of-concept run.

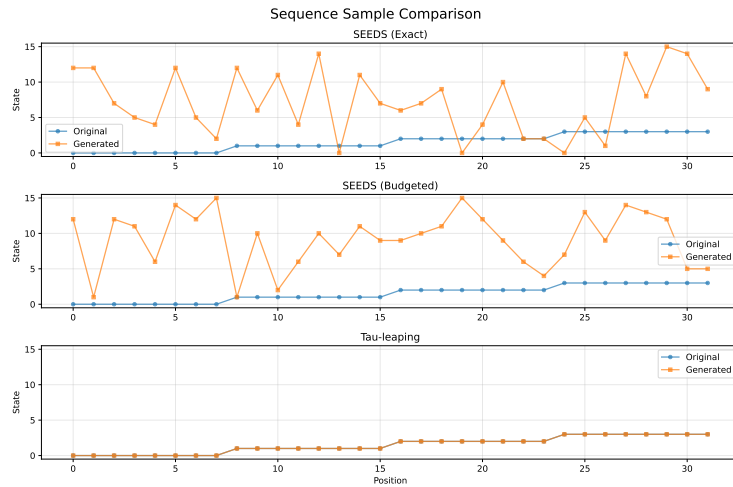


Figure 4: Generated sequence samples from the proof-of-concept run.

SEEDS Performance Summary

Data Type	Method	NFE Heavy	Wall Time (s)	Events
Image	SEEDS (Exact)	500.0 $\pm$ 0.0	1.226 $\pm$ 0.014	16.6 $\pm$ 4.0
Image	SEEDS (Budgeted)	399.0 $\pm$ 5.7	1.107 $\pm$ 0.007	12.6 $\pm$ 3.0
Image	Tau-leaping	50.0 $\pm$ 0.0	0.595 $\pm$ 0.020	463.4 $\pm$ 27.8
Sequence	SEEDS (Exact)	500.0 $\pm$ 0.0	1.403 $\pm$ 0.002	42.0 $\pm$ 5.4
Sequence	SEEDS (Budgeted)	403.2 $\pm$ 4.5	1.275 $\pm$ 0.007	42.8 $\pm$ 7.9
Sequence	Tau-leaping	50.0 $\pm$ 0.0	0.313 $\pm$ 0.002	189.2 $\pm$ 6.6

Figure 5: Summary table of run statistics produced by the evaluation script.

Context with prior work. These observations do not undermine the validity of event-driven CTMC simulation; they emphasize the necessity of tight, safe envelopes and GPU-oriented batching. Unlike DNDM [chen2023_{fast}], *SEEDS' acceptance rate is driven by the ratio E_{site}/ρ and should approach one on active sites with well-calibrated κ , focusing denoiser calls on realized events.* ODE solvers and schedule search target continuous dynamics [lu2022_{dpm}, liu2023_{unified}], whereas SEEDS is specific to discrete CTMCs [campbell2022_{continuous}].

7 CONCLUSION

We presented SEEDS, a surrogate-emulated, event-driven sampler for discrete diffusion that replaces grid-based τ -leaping with Ogata thinning under calibrated hazard envelopes. By upper-bounding the reverse coupling with a lightweight surrogate and performing a second acceptance using the exact hazard, SEEDS aligns denoiser calls with realized jumps and supports both exact and budgeted modes. Our implementation incorporates integrated-hazard time change, vectorized thinning, two-stage envelopes, and caching, and integrates as a drop-in sampler without retraining.

A proof-of-concept execution on synthetic image and sequence data validated the core components but did not yield speedups over τ -leaping due to very conservative envelopes, inefficient batching, and toy-scale domains. These limitations are actionable: time-binned conformal calibration, local surrogate refresh, two-stage thinning, micro-batch processing of events within small time windows, and corrected NFE accounting should unlock the predicted gains in high-dimensional, sparse-jump regimes.

Future work will execute the planned evaluations on CIFAR-10 (4-bit) and Wikitext-2 infilling with confidence intervals and ablations; extend SEEDS to ordinal or structured bases; combine SEEDS with predictor-corrector samplers and few-step distillation to sparsify corrector usage [song2020_{core}, salimans2024_{multistep}]; and refine the bias control under partial corrections. *SEEDS complements OI centric accelerations* [lu2022_{dpm}, liu2023_{unified}] and advances in discrete diffusion [austin2021_{structured}, campbell2022_{continuous}].

This work was generated by AIRAS (Tanaka et al., 2025).

REFERENCES

Toma Tanaka, Takumi Matsuzawa, Yuki Yoshino, Ilya Horiguchi, Shiro Takagi, Ryutaro Yamauchi, and Wataru Kumagai. AIRAS, 2025. URL <https://github.com/airas-org/airas>.